

Viyom Shukla

LinkedIn: viyom-shukla

GitHub: github.com/viyomshukla

TUF+: 350+ DSA problems | LeetCode: 200+ problems

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Portfolio: viyom-potfolio.vercel.app

Location: Noida, India

EDUCATION

- Jaypee Institute of Information Technology** Noida, India
Integrated B.Tech + M.Tech in Computer Science Engineering; CGPA: 7.62/10 2022 – 2027

EXPERIENCE

- Distance Connect** Remote
Full Stack Web Intern Aug 2025 – Oct 2025
 - Cross-Functional Product Development:** Collaborated directly with founders at an early-stage EdTech startup to redesign the platform UI, translating ambiguous, evolving requirements into a shipped product that improved student engagement.
 - Process Automation:** Engineered an AI-powered chatbot to automate student onboarding and query resolution, reducing manual support workload and improving turnaround time for user queries.
 - Third-Party API Integration:** Integrated the Google Meet API to build a secure, real-time 1:1 mentorship video pipeline, working independently to debug and deliver within a fast-paced startup environment.
- Student Placement Committee, IIIT Noida** Noida, India
Coordinator Aug 2024 – Jan 2025
 - Corporate Outreach:** Coordinated with the placement head and cross-functional student teams to source companies actively hiring for campus roles, compiling job requirements and eligibility criteria to prioritize outreach efforts.

PROJECTS

- Cybersecurity Anomaly Detection & Synthetic Data Pipeline (Python, TensorFlow, Scikit-Learn, XGBoost, Pandas)** Notebook
 - Data Processing & ETL Pipeline:** Designed an end-to-end ETL and anomaly detection pipeline on 100K+ transaction records, cleaning, joining, and engineering features to detect irregular activity patterns.
 - Feature Selection & Imbalance Correction:** Applied Recursive Feature Elimination (RFE) to isolate key predictive variables and implemented a Conditional GAN to address heavy class imbalance in rare transaction categories.
 - Ensemble Model Benchmarking:** Benchmarked XGBoost, Gradient Boosting, and Random Forest using ROC-AUC and confusion matrices to optimize anomaly classification and minimize false positives.
- DeepMood — GAN-Based Class-Imbalance Correction (Python, ML, Generative AI)** Notebook
 - Data Cleaning & Root Cause Analysis:** Cleaned and encoded a 414K-row behavioral dataset, then applied Random Forest-based Recursive Feature Elimination to isolate the key drivers behind a target outcome from 15+ candidate variables.
 - Class Imbalance Correction:** Diagnosed a 2:1 class imbalance skewing model predictions, then designed a Conditional GAN to synthesize tens of thousands of minority-class records and produce a more representative dataset for analysis.
 - Model Benchmarking & Insight Generation:** Benchmarked 9 classification models (Random Forest, Gradient Boosting, SVM, AdaBoost, Stacking, and others) using cross-validation, precision/recall/F1, and confusion matrices, surfacing a precision-recall trade-off with direct implications for real-world decision thresholds.
- Thumblify — AI-Powered Thumbnail Generation & Editing Platform (React.js, Node.js, MongoDB, REST APIs, Cloudinary)** Source Code
 - End-to-End Platform Architecture:** Built a full-stack SaaS (React.js, Node.js/Express) with fault-tolerant API integrations, serving 100+ users with zero downtime.
 - Auth & Payment Systems:** Implemented OAuth 2.0/JWT-secured APIs and a Razorpay-based credit subscription system with server-side webhook verification for all transactions.
 - Low-Latency Multi-Channel Access:** Built a Twilio WhatsApp bot and in-app chatbot for instant prompt-to-thumbnail generation, bypassing the web dashboard.

RESEARCH & PUBLICATIONS

- Enhancing Network Intrusion Detection Using Conditional GANs for Imbalanced Data Classification** IEEE Xplore
 - Implemented a Conditional GAN to synthesize rare attack samples in the NSL-KDD dataset, balancing class distribution and training Random Forest, SVM, and Neural Network classifiers to achieve 99.8% detection accuracy across underrepresented threat categories.

ACHIEVEMENTS

- SQL Certification:** HackerRank SQL Certified; solved 70+ SQL problems on LeetCode, focusing on query optimization and problem-solving.
- International Conference Speaker:** Presented published research at the 17th IEEE IC3-2025, IIIT Noida. *Certificate*
- National Semi-Finalist — Smart India Hackathon (SIH '25):** Selected among the top national teams across India in the Tourism domain.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C, C++

Data Analysis & Visualization: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Databases: MySQL, MongoDB

Core Concepts: Data Structures & Algorithms, DBMS, Machine Learning, Exploratory Data Analysis

Tools: Git, GitHub, VS Code, Jupyter Notebook, Postman, Notion, Slack